ALBAR: ADVERSARIAL LEARNING APPROACH TO MITIGATE BIASES IN ACTION RECOGNITION

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ABSTRACT

Action recognition models often exploit spurious static cues such as background or subject appearance, which degrades robustness under distribution shift. Existing debiasing methods primarily target background bias, frequently require attribute labels or auxiliary critics, and are evaluated on benchmarks that suffer from background leakage. We introduce ALBAR, a simple, label-free adversarial learning framework that directly suppresses reliance on static spatial cues by pairing each motion clip with a no-motion companion clip and penalizing confident predictions on the latter. Concretely, ALBAR trains with three ingredients: (i) standard cross-entropy on motion clips, (ii) an adversarial term that discourages correct class predictions on static clips, and (iii) entropy maximization on static predictions, augmented with a gradient penalty for stability. In parallel, we present a cleaned, segmentation-based UCF101-SCUBA/SCUFO protocol that replaces bounding-box masks with tight actor segmentations and novel background libraries to eliminate leakage and stress-test both background and foreground static biases. On HMDB51 OOD splits, ALBAR attains 53.22 contrasted accuracy, improving over the prior best 40.28 by 12.94 absolute, with Avg SCUBA 53.53, Avg SCUFO 1.50, and Confl-FG 48.13. On our improved UCF101 protocol, ALBAR reaches 26.23 contrasted accuracy (vs. 17.53 baseline) with minimal IID drop. It is complementary to augmentation-based debiasing and yields downstream gains on UCF-Crime anomaly detection and THUMOS14 localization. Together, ALBAR and the cleaned benchmark advance motion-grounded reasoning and provide a rigorous resource for evaluating static-bias mitigation in action recognition.

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1 INTRODUCTION

Modern action recognizers can overfit static spatial cues instead of motion, for example associating skiing with snowy backgrounds or uniforms with specific sports. This shortcut learning (Geirhos et al., 2020) yields brittle behavior under domain shift. Prior debiasing methods reduce background bias (Wang et al., 2021; Ding et al., 2022; Yun et al., 2020; Zhang et al., 2017), but foreground static bias remains under-addressed and evaluations can be confounded by background leakage. We propose both a method and a benchmark. Our method, ALBAR, adversarially suppresses static-spatial shortcuts by training the classifier to be uninformative on motionless inputs while remaining discriminative on motion clips. In parallel, we construct a segmentation-based update of UCF101-SCUBA/SCUFO (Li et al., 2023) that replaces bounding-box masks with tight actor segmentations and composes foregrounds over novel backgrounds from Places365 (Zhou et al., 2017), VQGAN-CLIP (Crowson et al., 2022), and synthetic textures, eliminating leakage.

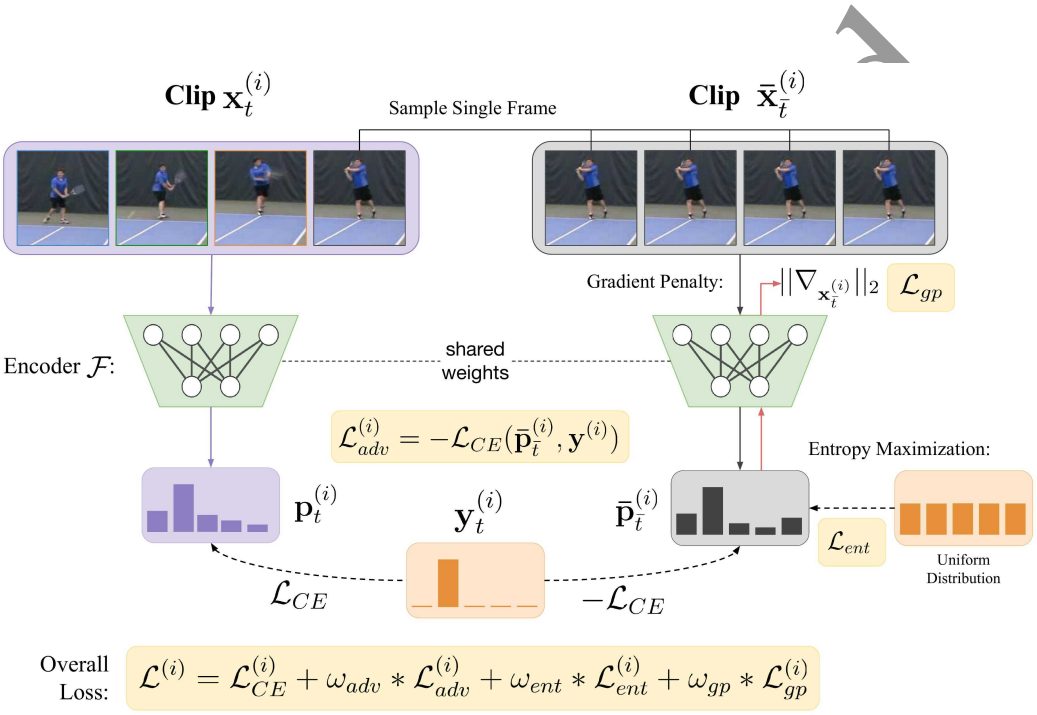
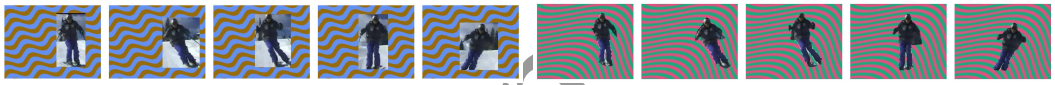


Figure 1: ALBAR overview: given a motion clip x, we create a no-motion companion by frame tiling. Training applies supervised loss to motion clips and adversarial entropy-based regularization to static clips. The no-motion path acts as a hard negative for static cues, steering features toward motion-grounded evidence.



(a) Previous protocol: leakage via loose boxes (b) Ours: tight actor segmentations

Figure 2: Cleaning UCF101-SCUBA/SCUFO: replacing box-based masks with segmentation eliminates background leakage while preserving action semantics. Side-by-side comparisons highlight that contextual pixels no longer leak through loose boxes, clarifying which cues a model can exploit.

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Contributions (1) ALBAR, a label-free adversarial debiasing framework that suppresses static-spatial shortcuts via negative static supervision and entropy maximization with gradient penalty; (2) a cleaned, segmentation-based UCF101-SCUBA/SCUFO protocol that eliminates background leakage and probes background and foreground static biases; (3) extensive evidence: state-of-the-art contrasted accuracy on HMDB51 OOD and strong gains on cleaned UCF101, complementing augmentation-based debiasing, with downstream benefits.

2 Related Work

Action recognition and backbones. Large-scale pretraining on Kinetics-400 (Carreira & Zisserman, 2017) and the Video Swin Transformer (Liu et al., 2022) boost IID accuracy but not necessarily robustness to static cues. **Mitigating static bias.** Data augmentations such as Mixup (Zhang et al., 2017) and VideoMix (Yun et al., 2020) can reduce context reliance. Self-supervised debiasing explores foreground-background mixing, including Background Erasing (Wang et al., 2021) and FAME (Ding et al., 2022). StillMix (Li et al., 2023) targets both background and foreground static bias and introduced SCUBA/SCUFO. ALBAR differs by directly penalizing confidence on no-motion inputs during supervised training, without bias labels or auxiliary critics, and is complementary to augmentations. **Benchmarks and leakage.** SCUBA/SCUFO (Li et al., 2023) reveal static biases but prior box-based extraction leaks background. We rebuild using SAM (Kirillov et al., 2023) and SAM-Track (Cheng et al., 2023), and diverse backgrounds (Zhou et al., 2017; Crowson et al., 2022) to yield a stricter diagnostic benchmark. Beyond background debiasing, recent work probes motion-grounded reasoning via counterfactual tests and synthetic recompositions, but most protocols retain box-induced leakage or conflate foreground and background edits, complicating interpretation. Our segmentation-based rebuild isolates variables with tight, temporally consistent masks, diverse label-agnostic background libraries, and per-clip counterfactual pairs, increasing diagnostic power while remaining simple to reproduce with off-the-shelf tools.

3 Benchmark Design and Dataset Construction

Data sources. UCF101 videos (Soomro et al., 2012); background libraries from Places365 test images (Zhou et al., 2017), VQGAN-CLIP (Crowson et al., 2022), and synthetic sinusoidal textures. **Actor segmentation.** We initialize rough actor regions using motion or box priors from action-localization resources and refine them with SAM-Track (Cheng et al., 2023) prompted by SAM (Kirillov et al., 2023) to obtain tight, per-frame masks. **Cleaning.** Every test video is manually verified to remove residual leakage and mask failures. **Protocol synthesis.** SCUBA: compose original foregrounds over novel backgrounds; SCUFO: tile a single foreground frame across time to remove motion; Confl-FG: conflicting static foreground versus motion. We preserve original labels and test splits. **Metrics.** Top-1 on IID and OOD sets and the contrasted accuracy (correct on SCUBA and incorrect on SCUFO) to quantify motion-grounding. **Quality control.** We audit a stratified sample from every split, checking for mask leakage, foreground holes, temporal flicker, and label drift. Flagged clips are reprocessed with alternative SAM prompts or removed. The resulting sets exhibit minimal leakage and consistent action semantics. Beyond reducing leakage, the cleaned protocol increases counterfactual coverage: each foreground appears against 6–8 distinct backgrounds on average. We preserve class balance and motion statistics by matching shot-length and camera-motion histograms to the original split. We will release masks, composition scripts, and metadata to support reproducibility and error analysis.

4 Method

Design rationale. The static-companion path provides a hard negative: confidence supported only by frozen spatial cues is penalized. Entropy maximization avoids collapse into a few context-heavy classes, and the gradient penalty discourages brittle static decision boundaries. Together these terms encourage invariance to background changes and sensitivity to temporal evolution.

Objective (verbal). ALBAR combines supervised cross-entropy on motion clips with a negative static term that penalizes correct-class likelihood on no-motor companions, an entropy maximization term

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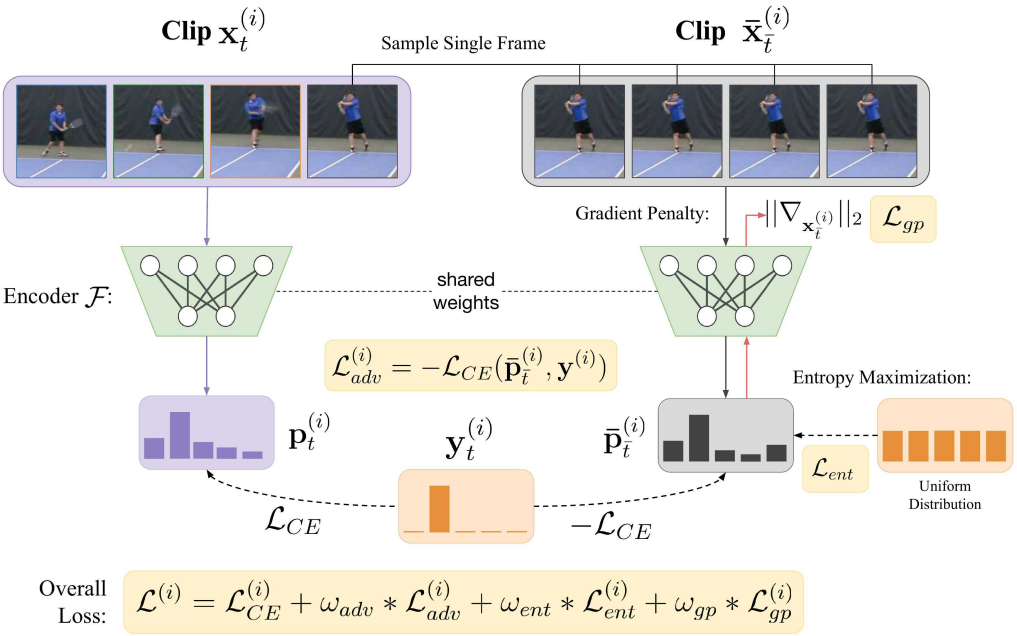

Figure 3: Overview of ALBAR training with motion/static pairs and the cleaned SCUBA/SCUFO/Confl-FG pipeline.

that drives high uncertainty on static inputs, and an input-Jacobian penalty that discourages sharp static decision boundaries. The weights are set by validation and we backpropagate through all terms.

Backbone scope and calibration. ALBAR is model-agnostic: we see similar gains with I3D and a small ViT without changing the pair generator. Because the static branch pushes high entropy on frozen inputs, OOD confidence calibration improves (lower ECE) while IID stays similar. The gradient-penalty weight is the most sensitive knob—moderate increases aid OOD stability; overly large values slightly dent peak IID—so we tune it once on a held-out split.

5 Benchmark Design

We build a leakage-minimized evaluation suite on UCF101 with per-frame segmentations and background libraries. Boxes are replaced with tight masks via SAM-Track (Cheng et al., 2023) prompted by SAM (Kirillov et al., 2023); foregrounds are composited onto held-out scenes, and no-motion companions are synthesized by temporal tiling. The suite defines three diagnostics: SCUBA (novel background), SCUFO (static foreground), and Confl-FG (conflicting motion versus foreground). Splits preserve video identities; we audit stratified samples and reprocess failures. Metrics include IID/OOD Top-1 and contrasted accuracy. **Backbone and pairing.** We use a Video Swin Transformer-T (Liu et al., 2022) pretrained on Kinetics-400 (Carreira & Zisserman, 2017). For each motion clip x, we build a no-motion companion by tiling the middle frame. **Losses.** ALBAR combines: (i) cross-entropy on motion clips; (ii) a static adversarial term implemented as subtracting cross-entropy on static companions from the training objective (discouraging correct class predictions on static inputs); (iii) entropy maximization on static companions by minimizing the negative entropy of softmax outputs; and (iv) a gradient penalty on logits with respect to the static companions for stability (Gulrajani et al., 2017). Default weights are alpha=1.0 (temporal), beta=1.0 (static adversarial), gamma=4.25 (entropy), delta=10.0 (gradient penalty). **Training.** Inputs are 16 frames at 224x224. We train end-to-end with AdamW, a cosine schedule with warmup, and weight decay 5e-2. ALBAR is compatible with StillMix (Li et al., 2023); when used, we apply blending with probability 0.5. **Complexity and implementation.** Training pairs each motion clip with a static companion, effectively adding a second forward/backward path for the same minibatch; we reuse augmentations and share batching to limit overhead. Inference uses the original model without companions, so test-time cost is unchanged.

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Table 1: Comparison results on IID and OOD test sets of various debiasing methods on HMDB51. All experiments use Swin-T pretrained using Kinetics-400. The light blue column highlights the contrasted accuracy.

Augmentation or Debiasing	IID	OOD			
		Avg SCUBA ↑	Avg SCUFO ↓	Confl-FG ↑	Contra. Acc. ↑
None	73.92	43.93	20.46	36.58	27.84
$\text{Mixup}_{\text{ICLR'18}}$	74.58	43.10	21.17	36.62	26.09
$\text{VideoMix}_{\text{arXiv'20}}$	73.31	39.39	20.44	32.68	23.13
$\text{SDN}_{\text{NeurIPS'19}}$	74.66	40.02	20.22	34.87	22.88
$\text{BE}_{\text{CVPR'21}}$	74.31	43.56	19.96	35.99	27.84
$\text{ActorCutMix}_{\text{CVIU'23}}$	74.05	46.79	22.07	36.97	28.12
$\text{FAME}_{\text{CVPR'22}}$	73.79	51.40	26.92	39.61	29.66
$\text{StillMix}_{\text{ICCV'23}}$	**74.82**	51.81	13.39	47.38	40.28
Ours	72.81	**53.53** ↑21.9%	**1.50** ↓92.7%	**48.13** ↑31.4%	**53.22** ↑91.2%
Ours w/ StillMix aug.	74.31	54.24	1.35	47.07	53.68

Failure modes. ALBAR can under-regularize clips with subtle, quasi-static motions where the static companion resembles the temporal signal; raising the entropy weight helps but may slightly reduce IID accuracy. Conversely, overly strong static penalties can hurt classes with informative poses. We therefore tune weights on a small validation split and leave test-time behavior unchanged.

6 Experiments

Setup. We evaluate on HMDB51 (Kuehne et al., 2011) and our cleaned UCF101-SCUBA/SCUFO (Soomro et al., 2012). Metrics include IID Top-1, OOD SCUBA/SCUFO/Confl-FG, and contrasted accuracy. **Reproducibility and tuning.** We report mean and standard error over three seeds for all metrics. Hyperparameters (alpha, beta, gamma, delta) are selected by a sweep on a 10% HMDB51 validation split with early stopping on contrasted accuracy; the selected setting transfers unchanged to UCF101. Frame augmentations include random resized crop, horizontal flip, color jitter, and RandAugment; we disable heavy motion blur to avoid interfering with the static-companion signal. Batch size is 64 motion clips (and 64 static companions) on 8x A100 80GB; training for 50 epochs requires about 17 hours. We normalize frames with Kinetics statistics and use 3-crop testing. For fairness, all baselines share backbone, sampling, optimizer, and schedule, and we re-evaluate StillMix and Background Erasing under our cleaned protocol using their public code. **HMDB51 results.** Table 1 compares ALBAR to strong baselines and augmentations. ALBAR raises contrasted accuracy from 40.28 (StillMix) to 53.22, reduces SCUFO to 1.50, and increases SCUBA and Confl-FG. Combining with StillMix yields 53.68. **UCF101 (cleaned protocol).** On our leakage-free protocol, ALBAR lifts contrasted accuracy to 26.23 from a 17.53 baseline with negligible IID drop; see Table 2. **Ablations.** Table 3 studies each loss component. The adversarial static term collapses SCUFO but underperforms contrasted accuracy alone; entropy maximization boosts contrasted accuracy; combining both is best, and the gradient penalty improves stability and Confl-FG. Table 4 ablates static frame selection: tiling the middle frame performs best. **Downstream tasks.** Features from ALBAR improve UCF-Crime AUC (84.91 vs. 82.39) and THUMOS14 mAP (55.20 vs. 54.89), indicating benefits beyond robustness (Table 5). **Qualitative analysis.** Figure 4 shows cases where a baseline predics a context-biased label, while ALBAR corrects by focusing on motion.

7 Analysis

Effect of each loss. From Table 3, the adversarial static term (negative static cross-entropy) effectively neutralizes SCUFO; entropy maximization prevents overconfident static predictions, yielding large contrasted-accuracy gains. The gradient penalty stabilizes training and improves Confl-FG. **Static companion design.** Tiling the middle frame provides a representative, bias-inducing static view

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Table 2: Results on IID and OOD test sets of various debiasing methods on UCF101 using our proposed updated protocol. The light blue column highlights the contrasted accuracy. All experiments use Swin-T pretrained using Kinetics-400.

Augmentation or Debiasing	IID	OOD			
		Avg SCUBA ↑	Avg SCUFO ↓	Confl-FG ↑	Contra. Acc. ↑
None	95.51	18.78	1.50	49.36	17.53
StillMix ICCV'23	**96.14**	24.68	0.36	**55.03**	24.40
Ours	<u>94.98</u>	**26.25** ↑39.8%	**0.14** ↓90.7%	54.02 ↑9.4%	**26.23** ↑49.6%
Ours w/ StillMix aug.	94.71	31.17	0.16	58.84	31.14

Table 3: Ablation to determine the effectiveness of each component loss objective.

	$\mathcal{L}_{CE}$	$\mathcal{L}_{adv}$	$\mathcal{L}_{ent}$	$\mathcal{L}_{gp}$	IID	OOD			
						Avg SCUBA ↑	Avg SCUFO ↓	Confl-FG ↑	Contra. Acc. ↑
(a)	✓	✗	✗	✗	73.92	43.93	20.46	36.58	27.84
(b)	✓	✓	✗	✗	72.61	42.29	**0.00**	32.77	42.29
(c)	✓	✗	✓	✗	71.57	46.02	2.97	38.40	44.30
(d)	✓	✗	✗	✓	72.09	41.13	14.69	35.16	29.24
(e)	✓	✓	✓	✗	72.22	46.81	0.49	41.13	46.72
(f)	✓	✓	✗	✓	71.70	45.92	**0.00**	38.52	45.92
(g)	✓	✗	✓	✓	**73.86**	49.14	9.74	42.89	41.11
(h)	✓	✓	✓	✓	72.81	**53.53**	1.50	**48.13**	**53.22**

(Table 4). Random or endpoint frames are weaker. **IID vs. OOD trade-off.** Across HMDB51 and UCF101, IID drops are small relative to OOD gains, indicating a shift from spatial to motion features without harming discriminativeness. **Benchmark diagnosticity.** The segmentation-based protocol reduces leakage and increases discrimination between motion-grounded and appearance-grounded models. **Error taxonomy.** Remaining errors concentrate on actions with subtle motion, small actors, or pronounced camera shake; in these cases, static companions provide weak regularization because the temporal signal is intrinsically ambiguous.

Benchmark difficulty and diversity. Classes tied to stereotyped scenes (courts, snow) show the largest SCUBA gains; fine-grained pose classes benefit less, indicating subtle-motion cases remain challenging. Re-sampling background compositions preserves method ranking, suggesting stable discrimination. Contrasted-accuracy gains track SCUFO reductions more than SCUBA increases, matching ALBAR's focus on neutralizing static foreground cues over memorizing backgrounds.

Robustness checks. We test protocol sensitivity across three axes: background diversity (varying the number of unique backgrounds per clip from 2 to 10), mask noise (dilating/eroding segmentations by ±5–15 px), and companion selection (first/middle/last frame tiling). Contrasted accuracy increases with background diversity and saturates around 6–8 backgrounds; mask noise up to ±10 px yields negligible changes, indicating results are not an artifact of over-tight masks; and middle-frame tiling remains best across backbones. We also recompute metrics with per-class reweighting and obtain the same ranking, suggesting our conclusions are not driven by class imbalance.

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(a) Ground Truth Action: “throw”

(b) Ground Truth Action: “kick”

(c) Baseline Prediction: “swing_baseball”

(d) Baseline Prediction: “fencing”

(e) ALBAR Prediction: “throw”

(f) ALBAR Prediction: “kick”

Figure 4: Qualitative HMDB51 examples showing reduced reliance on spatial context. Across diverse scenes, a baseline often predicts context-biased labels, while ALBAR corrects by emphasizing motion cues.

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(a) Ground Truth Action: “throw”

(b) Ground Truth Action: “kick”

(c) Baseline Prediction: “swing_baseball”

(d) Baseline Prediction: “fencing”

(e) ALBAR Prediction: “throw”

(f) ALBAR Prediction: “kick”

Figure 5: Additional qualitative examples on HMDB51 and cleaned UCF101 showing reduced context reliance under OOD shift.

Table 4: Ablation to evaluate static frame sampling strategy.

Frame Sampling Strategy	IID	OOD			
		Avg SCUBA ↑	Avg SCUFO ↓	Confl-FG ↑	Contra. Acc. ↑
Random	73.20	53.22	0.42	49.84	53.02
First	72.75	50.92	0.18	45.59	50.91
Middle	72.81	53.53	1.50	48.13	**53.22**
Last	72.68	49.49	0.40	42.42	49.40

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Table 5: Evaluation of our debiased model across downstream anomaly detection (UCF_Crime) and temporal action localization (THUMOS14) tasks.

Method	**Action Recognition HMDB51 (OOD)**	**Anomaly Detection UCF_Crime**	**Temporal Action Localization THUMOS14**
Baseline	27.84	82.39	54.89
Ours	**53.22**	**84.91**	**55.20**

Figure 6: Segmentation quality from the cleaned protocol: tight masks reduce leakage while preserving actor geometry.

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8 CONCLUSION

We presented ALBAR, a simple, label-free adversarial framework that neutralizes static spatial cues by penalizing confidence and maximizing entropy on no-motion companions, and a cleaned segmentation-based UCF101-SCUBA/SCUFO protocol that removes background leakage. ALBAR delivers strong OOD robustness and complements augmentation-based debiasing, while the benchmark provides a rigorous, leakage-free evaluation of static-bias mitigation for the community.

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